



CMPT 825: Natural Language Processing

Dependency Parsing

Spring 2020
2020-03-26

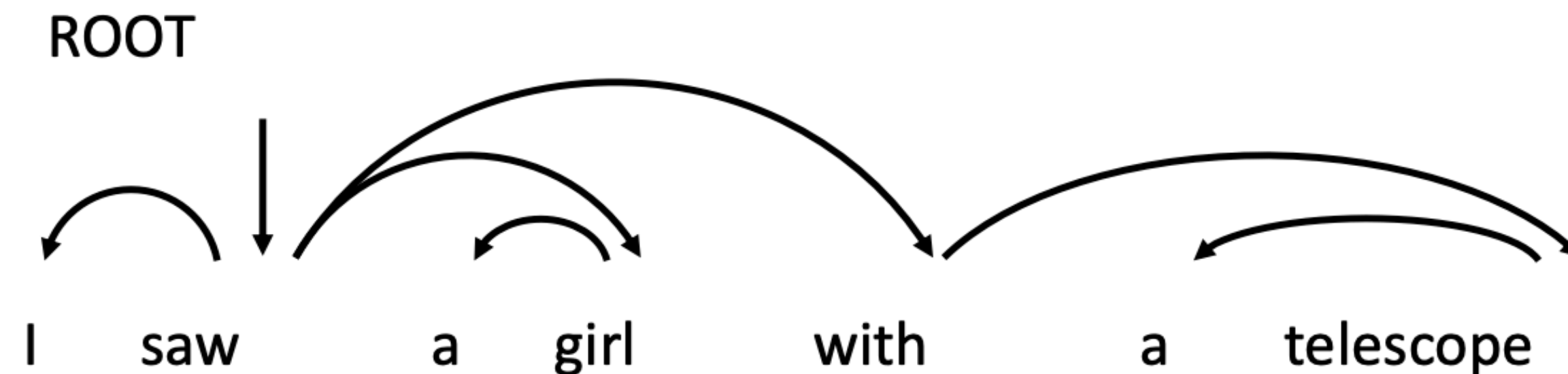
Adapted from slides from Danqi Chen and Karthik Narasimhan
(with some content from slides from Chris Manning and Graham Neubig)

Overview

- What is dependency parsing?
- Two families of algorithms
 - Transition-based dependency parsing
 - Graph-based dependency parsing

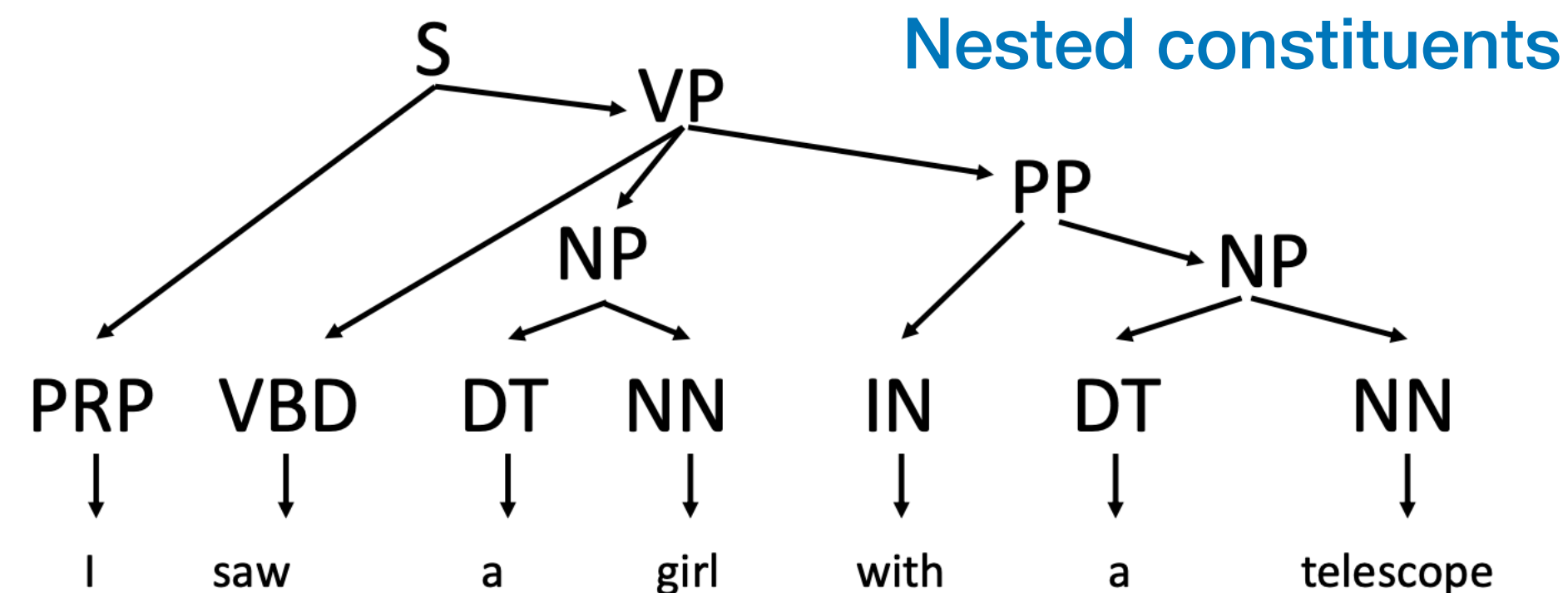
Dependency and constituency

- **Dependency Trees** focus on relations between words



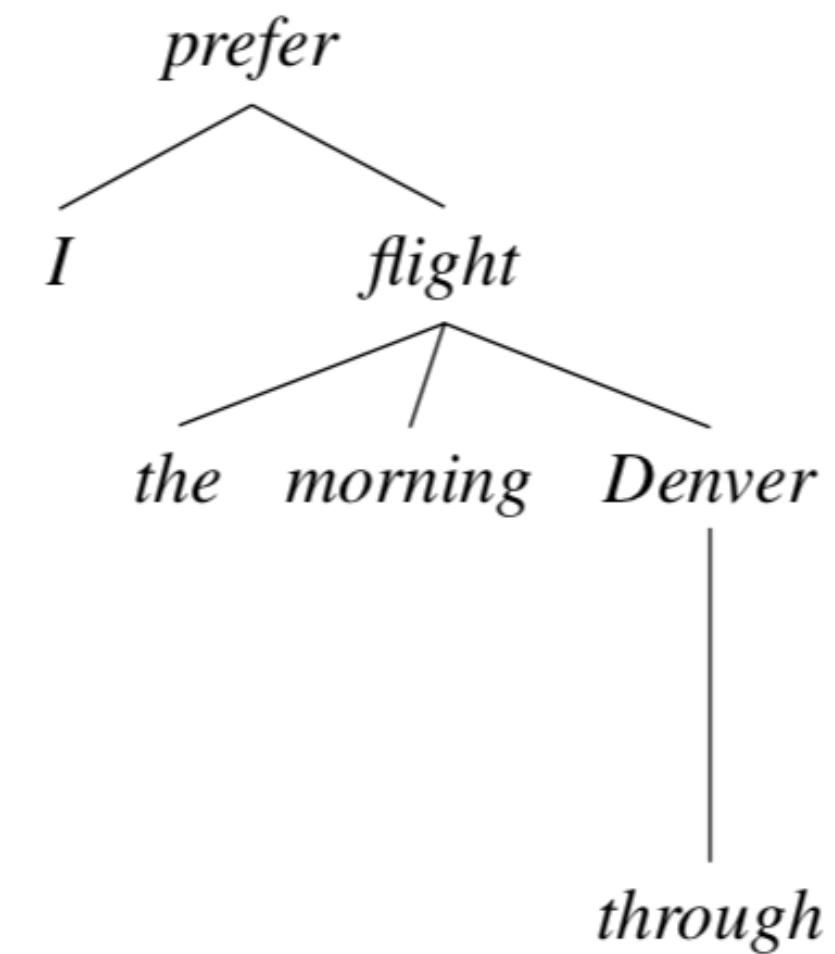
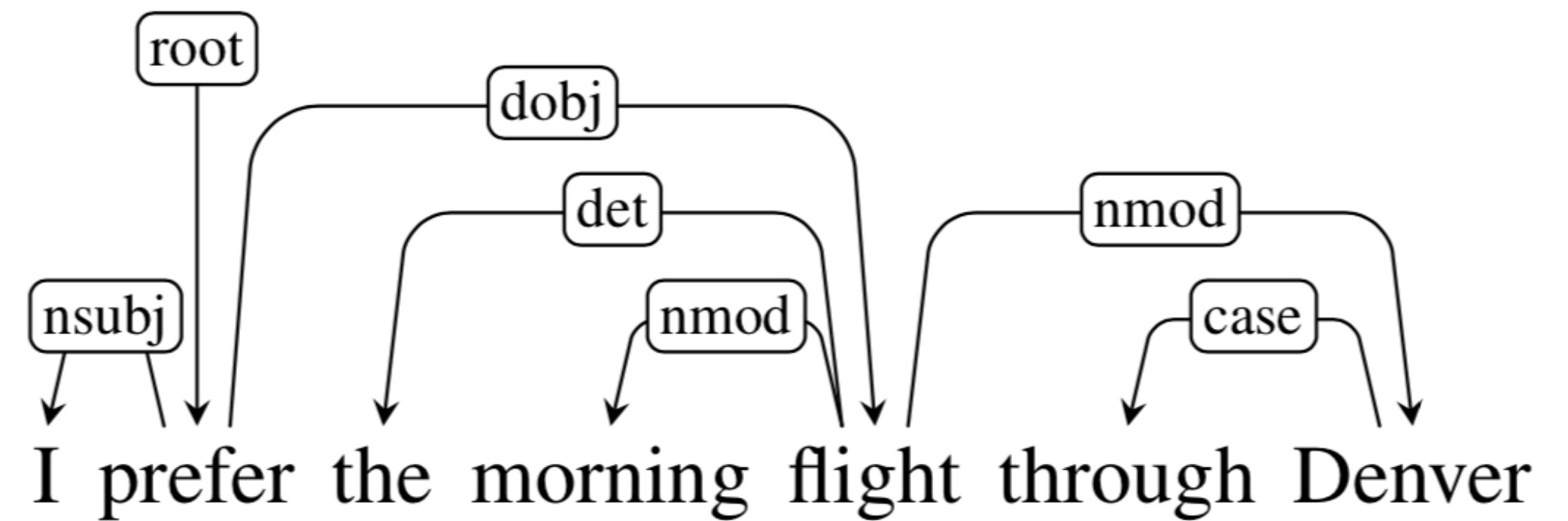
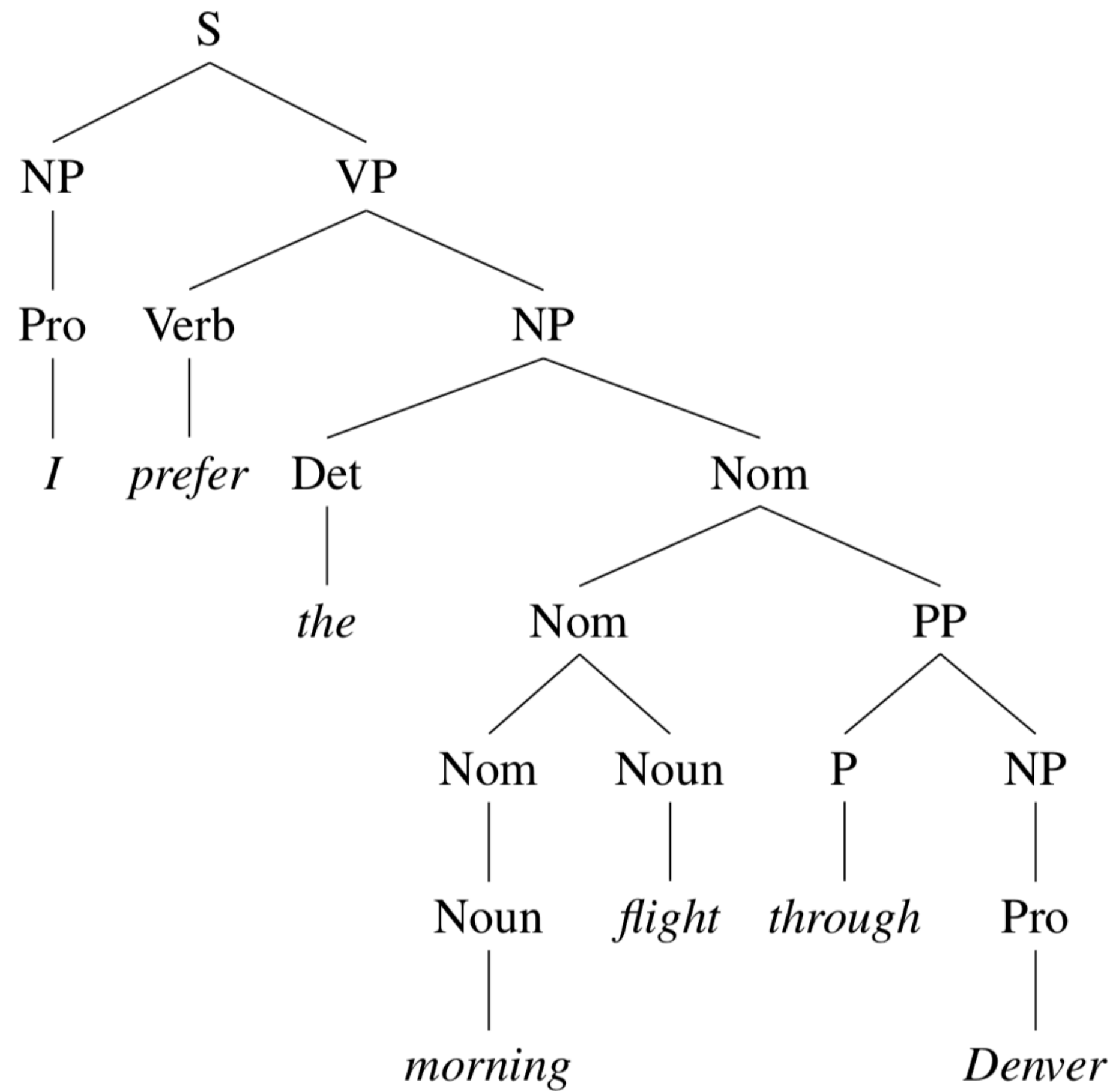
Words directly linked to each other

- **Phrase Structure** models the structure of a sentence



Constituency Parse
generated from
Context Free Grammars
(CFGs)

Constituency vs dependency structure



Pāṇini's grammar of Sanskrit (c. 5th century BCE)



Gallery: <http://wellcomeimages.org/indexplus/image/L0032691.html>

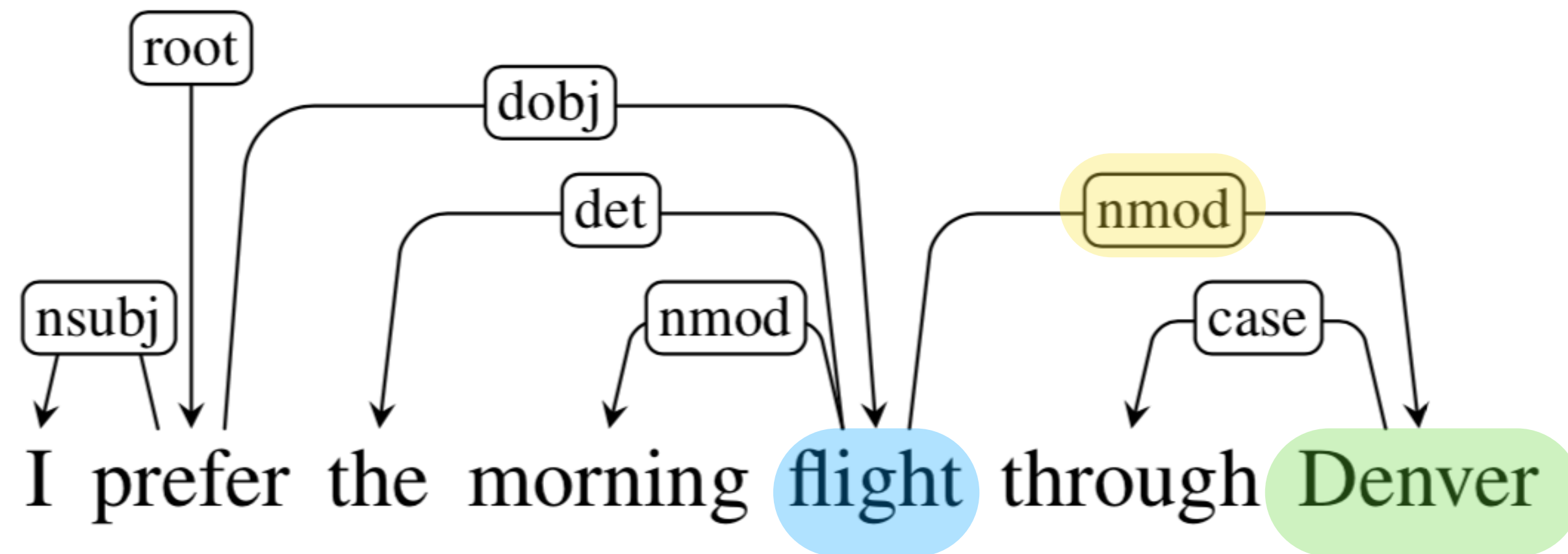
CC BY 4.0 File: Birch bark MS from Kashmir of the Rupavatra Wellcome L0032691.jpg

(slide credit: Stanford CS224N, Chris Manning)

Dependency Grammar/Parsing History

- The idea of dependency structure goes back a long way
 - To Pāṇini's grammar (c. 5th century BCE)
 - Basic approach of 1st millennium Arabic grammarians
- Constituency/context-free grammars is a new-fangled invention
 - 20th century invention (R.S. Wells, 1947; then Chomsky)
- Modern dependency work often sourced to L. Tesnière (1959)
 - Was dominant approach in “East” in 20th Century (Russia, China, ...)
 - Good for free-er word order languages
- Among the earliest kinds of parsers in NLP, even in the US:
 - David Hays, one of the founders of U.S. computational linguistics, built early (first?) dependency parser (Hays 1962)

Dependency structure



- Consists of relations between lexical items, normally *binary*, *asymmetric* relations (“arrows”) called **dependencies**
- The arrows are commonly typed with the name of grammatical **relations** (subject, prepositional object, apposition, etc)
- The arrow connects a **head** (governor) and a **dependent** (modifier)
- Usually, dependencies form a tree (single-head, connected, acyclic)

Dependency relations

Clausal Argument Relations	Description
NSUBJ	Nominal subject
DOBJ	Direct object
IOBJ	Indirect object
CCOMP	Clausal complement
XCOMP	Open clausal complement
Nominal Modifier Relations	Description
NMOD	Nominal modifier
AMOD	Adjectival modifier
NUMMOD	Numeric modifier
APPOS	Appositional modifier
DET	Determiner
CASE	Prepositions, postpositions and other case markers
Other Notable Relations	Description
CONJ	Conjunct
CC	Coordinating conjunction

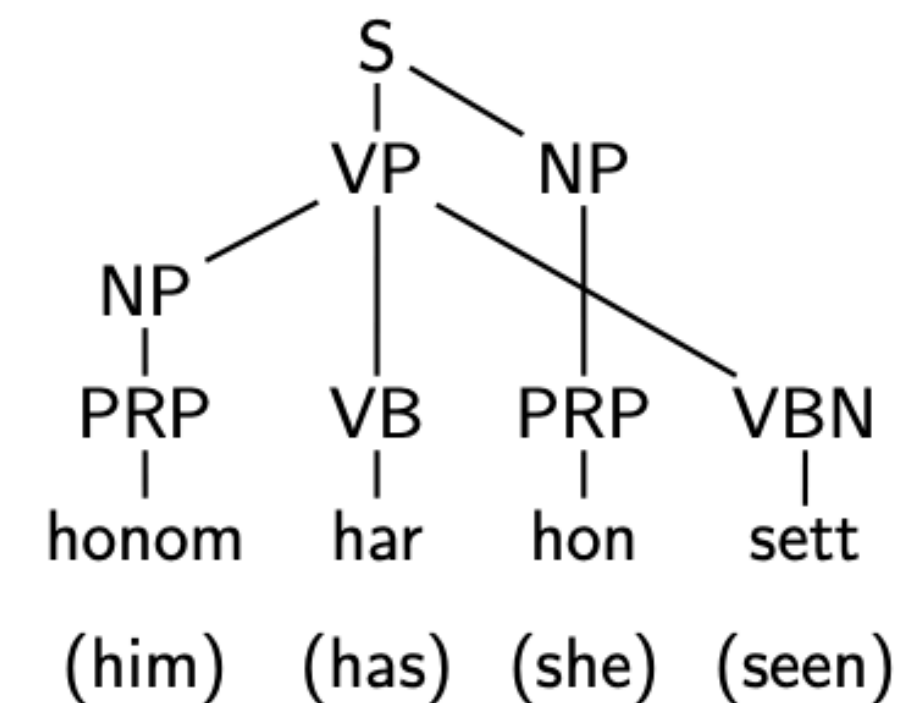
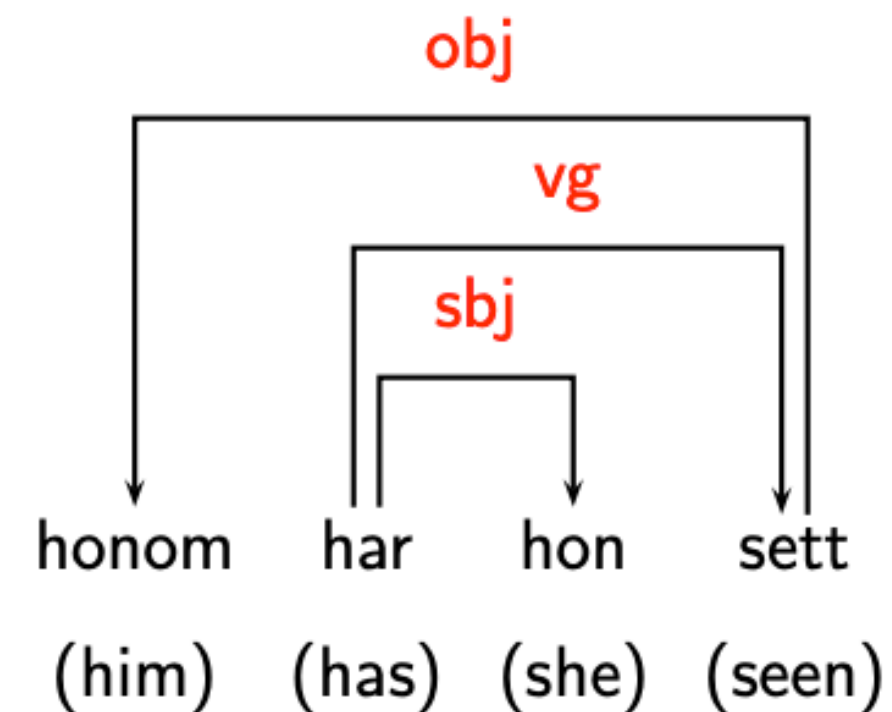
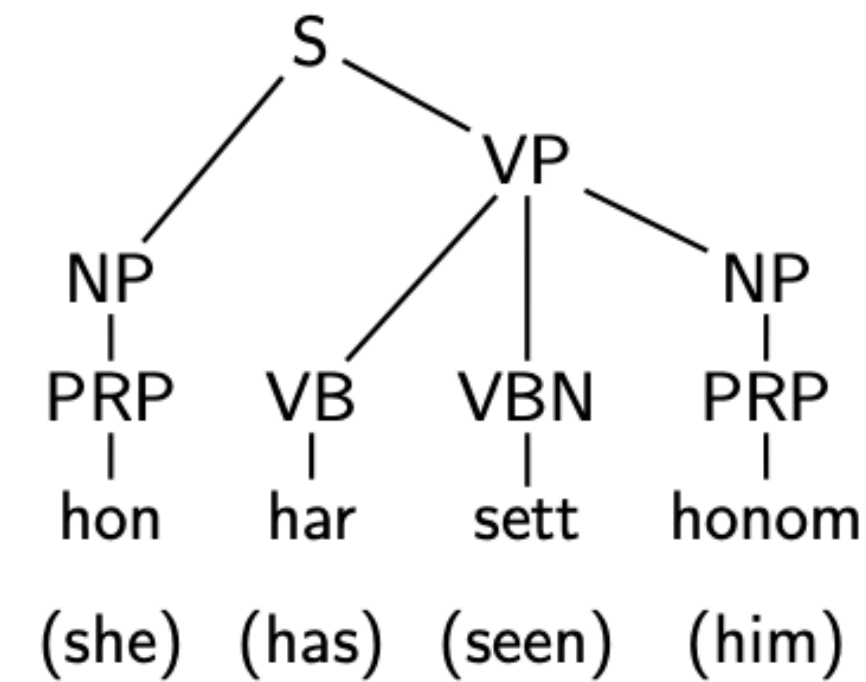
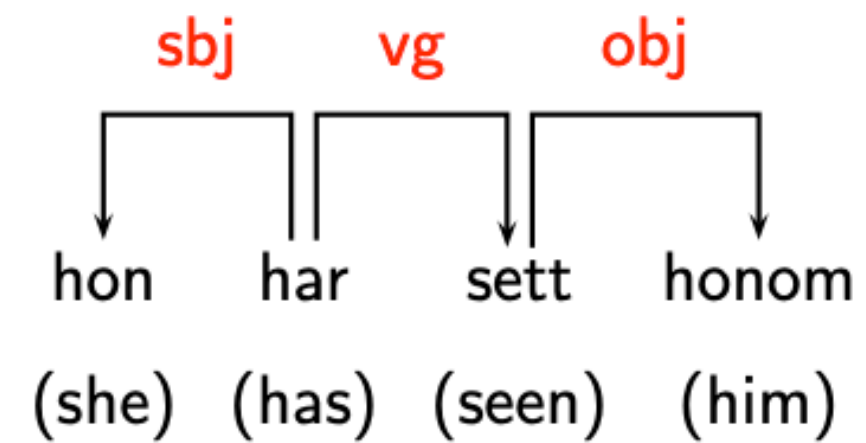
(de Marneffe and Manning, 2008): Stanford typed dependencies manual

Dependency relations

Relation	Examples with <i>head</i> and dependent
NSUBJ	United <i>canceled</i> the flight.
DOBJ	United <i>diverted</i> the flight to Reno. We <i>booked</i> her the first flight to Miami.
IOBJ	We <i>booked</i> her the flight to Miami.
NMOD	We took the morning <i>flight</i> .
AMOD	Book the cheapest <i>flight</i> .
NUMMOD	Before the storm JetBlue canceled 1000 <i>flights</i> .
APPOS	<i>United</i> , a unit of UAL, matched the fares.
DET	The <i>flight</i> was canceled. Which <i>flight</i> was delayed?
CONJ	We <i>flew</i> to Denver and drove to Steamboat.
CC	We flew to Denver and <i>drove</i> to Steamboat.
CASE	Book the flight through <i>Houston</i> .

Advantages of dependency structure

- More suitable for free word order languages



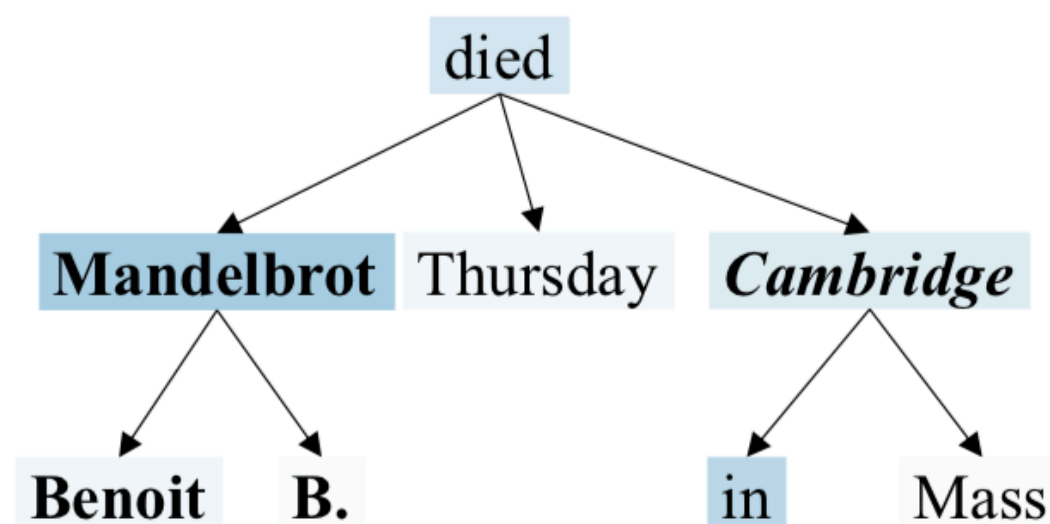
Advantages of dependency structure

- More suitable for free word order languages
- The predicate-argument structure is more useful for many applications

Relation Extraction

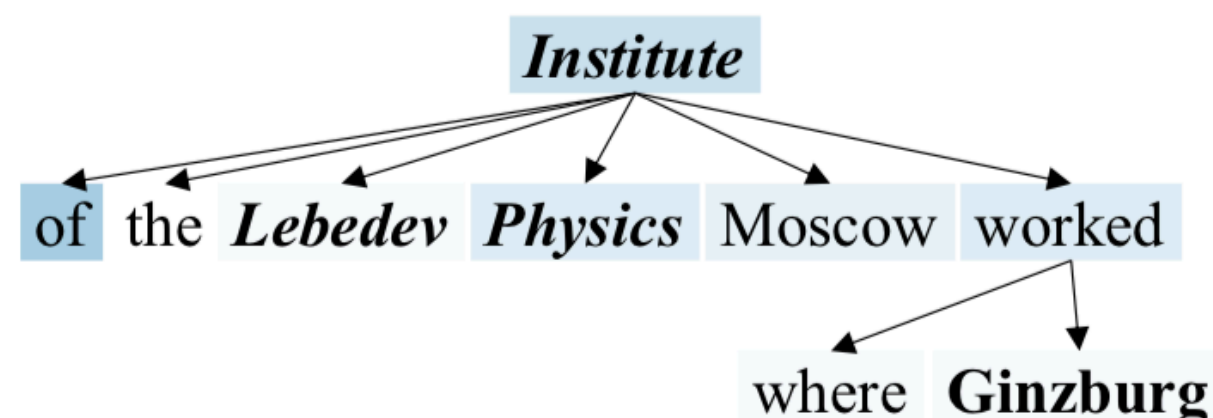
Relation: *per:city_of_death*

Benoit B. Mandelbrot, a maverick mathematician who developed an innovative theory of roughness and applied it to physics, biology, finance and many other fields, died Thursday in **Cambridge**, Mass.



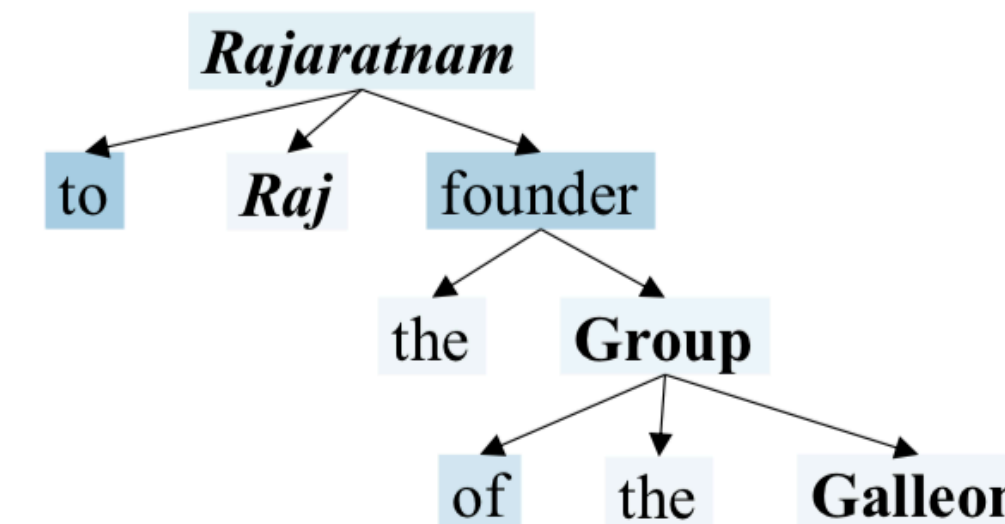
Relation: *per:employee_of*

In a career that spanned seven decades, Ginzburg authored several groundbreaking studies in various fields -- such as quantum theory, astrophysics, radio-astronomy and diffusion of cosmic radiation in the Earth's atmosphere -- that were of "Nobel Prize caliber," said Gennady Mesyats, the director of the **Lebedev Physics Institute** in Moscow, where **Ginzburg** worked .



Relation: *org:founded_by*

Anil Kumar, a former director at the consulting firm McKinsey & Co, pleaded guilty on Thursday to providing inside information to **Raj Rajaratnam**, the founder of the **Galleon Group**, in exchange for payments of at least \$ 175 million from 2004 through 2009.

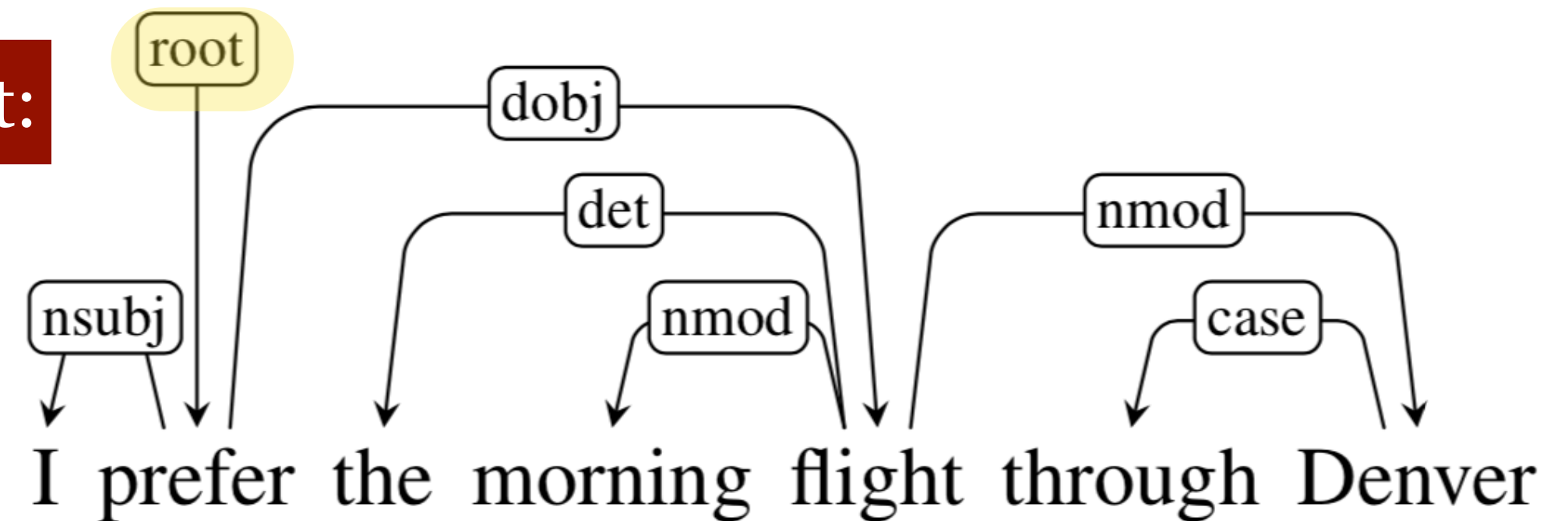


Dependency parsing

Input:

I prefer the morning flight
through Denver

Output:



- A sentence is parsed by choosing for each word what other word is it a dependent of (and also the relation type)
- We usually add a **fake ROOT** at the beginning so every word has one head
- Usually some constraints:
 - Only one word is a dependent of ROOT
 - No cycles: $A \rightarrow B, B \rightarrow C, C \rightarrow A$

Learning from data: treebanks!

Dependency Conditioning Preferences

What are the sources of information for dependency parsing?

1. Bilexical affinities [discussion → issues] is plausible
2. Dependency distance mostly with nearby words
3. Intervening material

Dependencies rarely span intervening verbs or punctuation

4. Valency of heads

How many dependents on which side are usual for a head?



Dependency treebanks

- The major English dependency treebank: converting from Penn Treebank using rule-based algorithms

Stanford
Dependencies
(English)

- (De Marneffe et al, 2006): Generating typed dependency parses from phrase structure parses
- (Johansson and Nugues, 2007): Extended Constituent-to-dependency Conversion for English

- Universal Dependencies: more than 100 treebanks in 70 languages were collected since 2016

Universal
Dependencies
(Multilingual)



Universal Dependencies (UD) is a framework for consistent annotation of grammar (parts of speech, morphological features, and syntactic dependencies) across different human languages. UD is an open community effort with over 200 contributors producing more than 100 treebanks in over 70 languages. If you're new to UD, you should start by reading the first part of the Short Introduction and then browsing the annotation guidelines.

<https://universaldependencies.org/>

Universal Dependencies

▶		Afrikaans	1	49K	↕	IE, Germanic
▶		Akkadian	1	1K	📖	Afro-Asiatic, Semitic
▶		Amharic	1	10K	☁️📖📝	Afro-Asiatic, Semitic
▶		Ancient Greek	2	416K	☁️📖	IE, Greek
▶		Arabic	3	1,042K	📖W	Afro-Asiatic, Semitic
▶		Armenian	1	36K	📖📝	IE, Armenian
▶		Assyrian	1	<1K	📖	Afro-Asiatic, Semitic
▶		Bambara	1	13K	📖	Mande
▶		Basque	1	121K	📖	Basque
▶		Belarusian	1	13K	📖↕📖	IE, Slavic
▶		Breton	1	10K	📖📝📖W	IE, Celtic
▶		Bulgarian	1	156K	📖↕	IE, Slavic
▶		Buryat	1	10K	📖📝	Mongolic
▶		Cantonese	1	13K	🗨️	Sino-Tibetan
▶		Catalan	1	531K	📖	IE, Romance
▶		Chinese	5	161K	📖📖📖🗨️W	Sino-Tibetan
▶		Classical Chinese	1	55K	📖	Sino-Tibetan
▶		Coptic	1	25K	☁️📖	Afro-Asiatic, Egyptian
▶		Croatian	1	199K	📖📖W	IE, Slavic
▶		Czech	5	2,222K	📖↕📖📖W	IE, Slavic
▶		Danish	2	100K	📖📖🗨️	IE, Germanic
▶		Dutch	2	307K	📖W	IE, Germanic
▶		English	6	603K	📖📖📖📖📖📖W	IE, Germanic
▶		Erzya	1	15K	📖	Uralic, Mordvin
▶		Estonian	2	461K	📖📖📖📖	Uralic, Finnic
▶		Faroese	1	10K	W	IE, Germanic
▶		Finnish	3	377K	📖📖📖W	Uralic, Finnic
▶		French	8	1,156K	📖↕📖📖🗨️W	IE, Romance
▶		Galician	2	164K	↕📖	IE, Romance
▶		German	4	3,409K	📖📖📖W	IE, Germanic
▶		Gothic	1	55K	☁️	IE, Germanic
▶		Greek	1	63K	📖🗨️W	IE, Greek
▶		Hebrew	1	161K	📖	Afro-Asiatic, Semitic
▶		Hindi	2	375K	📖W	IE, Indic
▶		Hindi English	1	26K	📖	Code switching
▶		Hungarian	1	42K	📖	Uralic, Ugric
▶		Indonesian	2	141K	📖📖W	Austronesian, Malayo-Sumbawan
▶		Irish	1	23K	📖↕📖	IE, Celtic
▶		Italian	6	781K	↕📖📖W	IE, Romance
▶		Japanese	5	1,688K	📖📖📖📖W	Japanese
▶		Karelian	1	3K	📖📖	Uralic, Finnic
▶		Kazakh	1	10K	📖W	Turkic, Northwestern
▶		Komi Zyrian	2	3K	📖🗨️	Uralic, Permian
▶		Korean	5	446K	📖📖📖📖W	Korean

Universal Dependencies

- Developing cross-linguistically consistent treebank annotation for many languages
- Goals:
 - Facilitating multilingual parser development
 - Cross-lingual learning
 - Parsing research from a language typology perspective.

Universal Dependencies



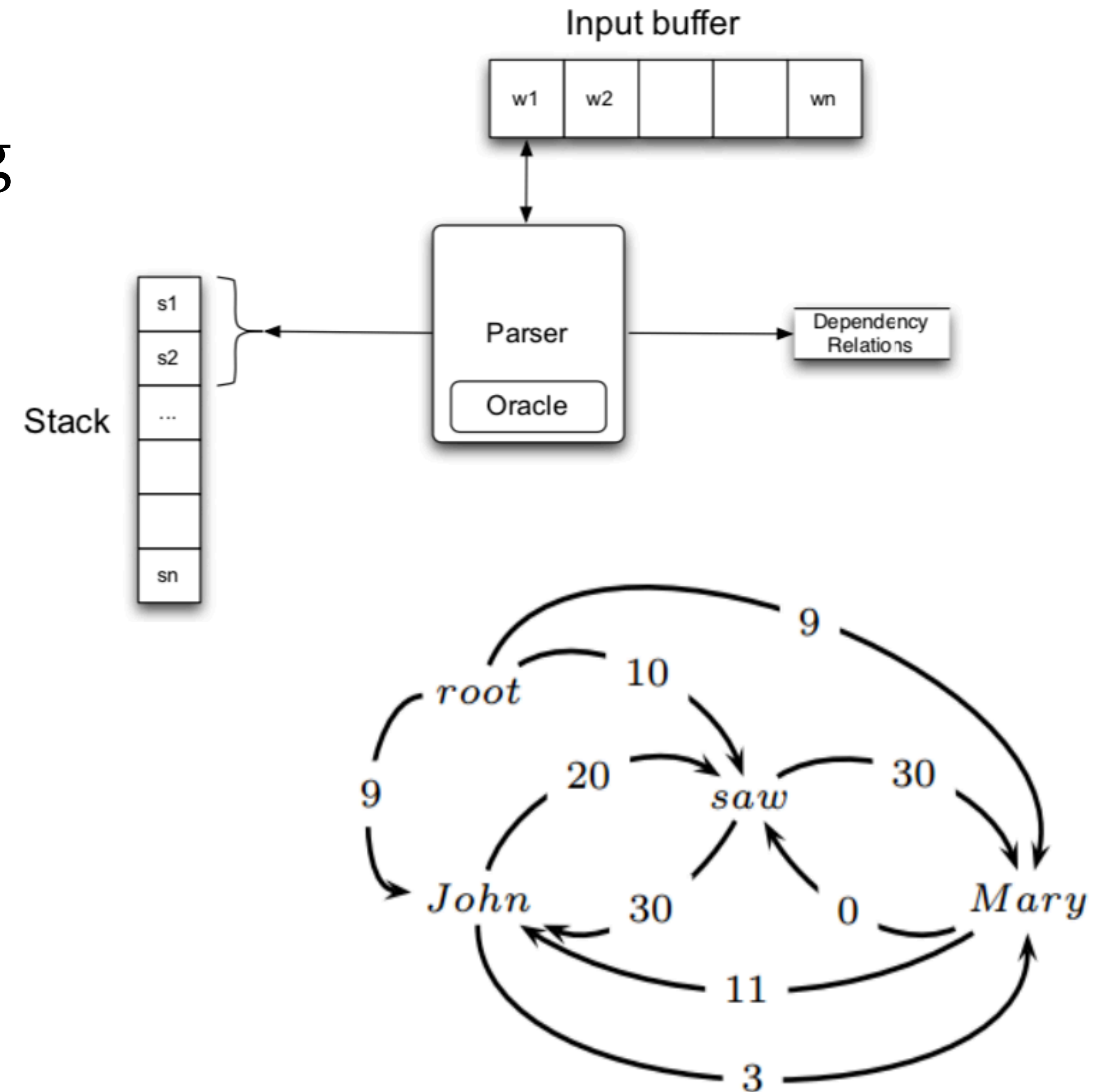
Manning's Law:

- UD needs to be satisfactory for analysis of individual languages.
- UD needs to be good for linguistic typology.
- UD must be suitable for rapid, consistent annotation.
- UD must be suitable for computer parsing with high accuracy.
- UD must be easily comprehended and used by a non-linguist.
- UD must provide good support for downstream NLP tasks.

Two families of algorithms

Transition-based dependency parsing

- Also called “shift-reduce parsing”



Graph-based dependency parsing

Two families of algorithms

Transition-Based

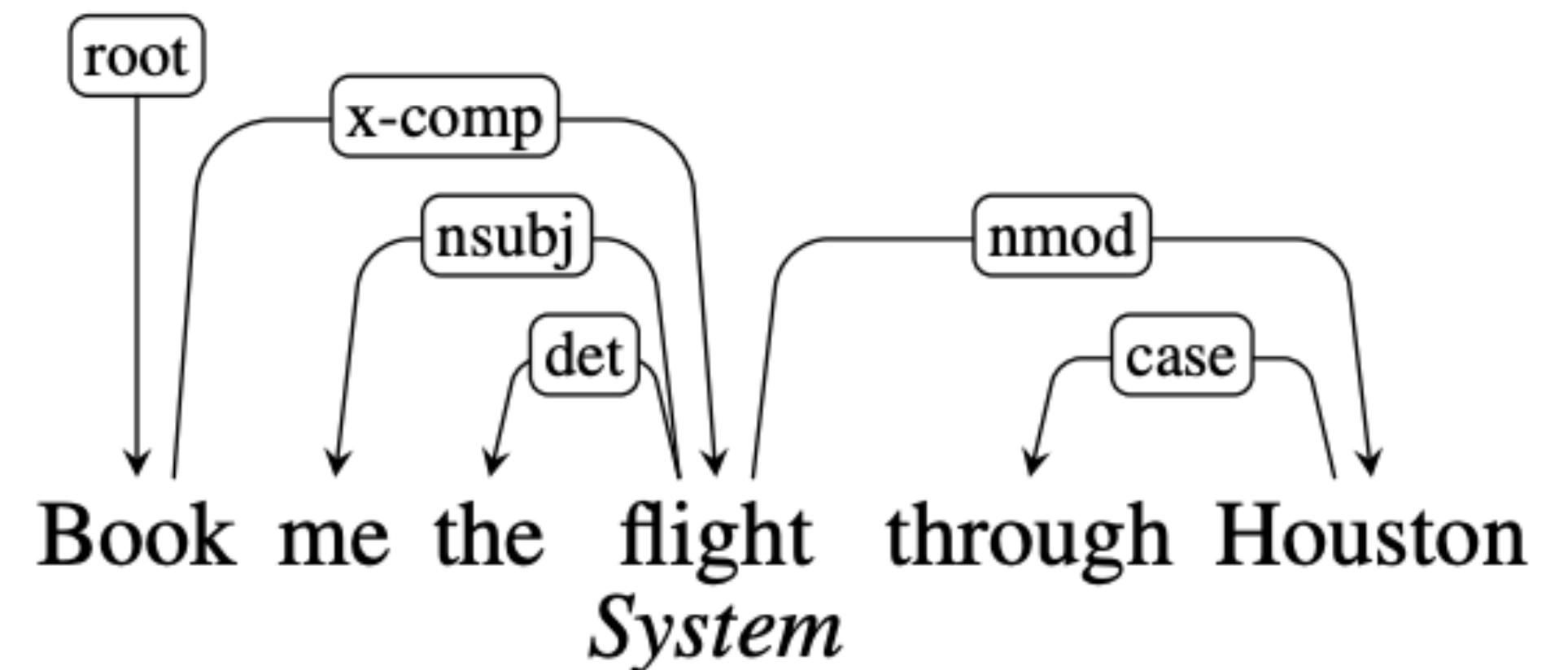
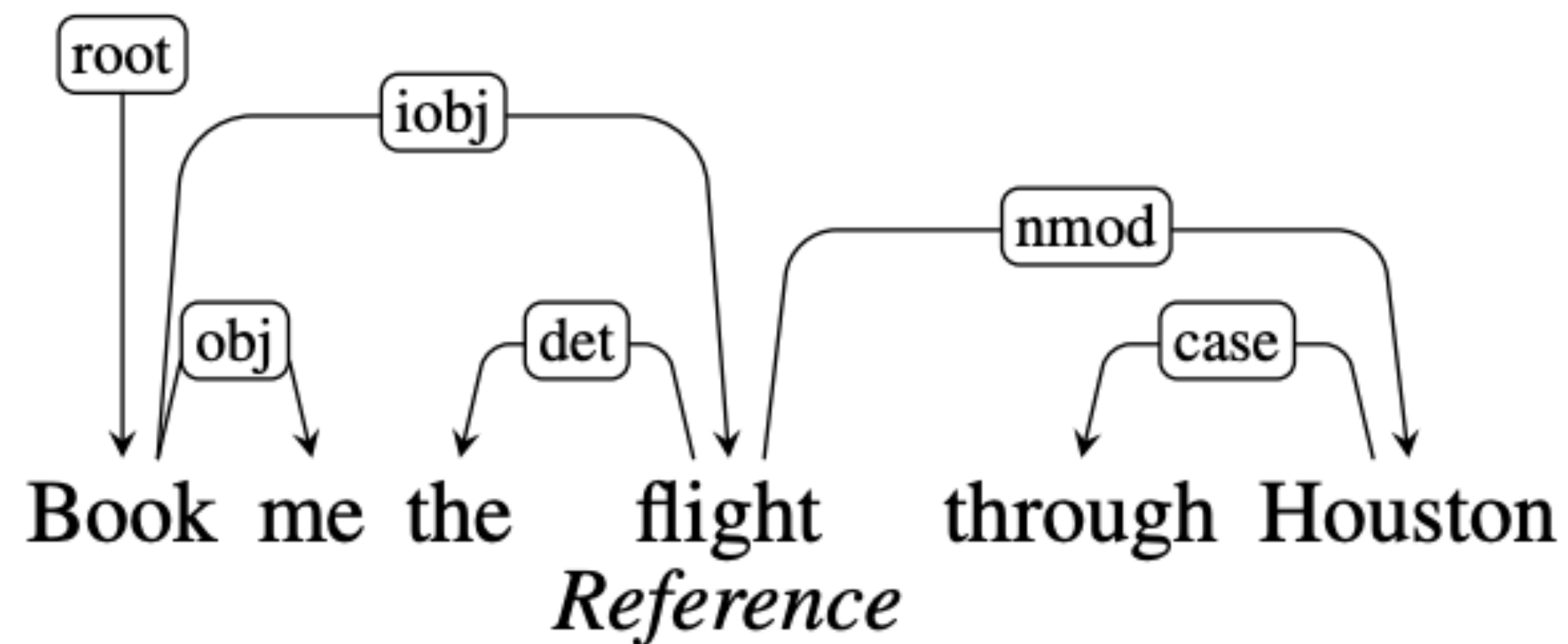
Parser		Test	
		UAS	LAS
(Chen and Manning, 2014)	T	91.8	89.6
(Dyer et al., 2015)		93.1	90.9
(Ballesteros et al., 2016)		93.56	92.41
(Weiss et al., 2015)		94.26	91.42
(Andor et al., 2016)		94.61	92.79
(Ma et al., 2018) §		95.87	94.19
(Kiperwasser and Goldberg, 2016a) §	G	93.0	90.9
(Kiperwasser and Goldberg, 2016b)		93.1	91.0
(Wang and Chang, 2016)		94.08	91.82
(Cheng et al., 2016)		94.10	91.49
(Kuncoro et al., 2016)		94.26	92.06
(Zheng, 2017) §		95.53	93.94
(Dozat and Manning, 2017)		95.74	94.08
Baseline	G	95.68	93.96
Our Model §		95.97	94.31

Graph-Based

T: transition-based / G: graph-based

Evaluation

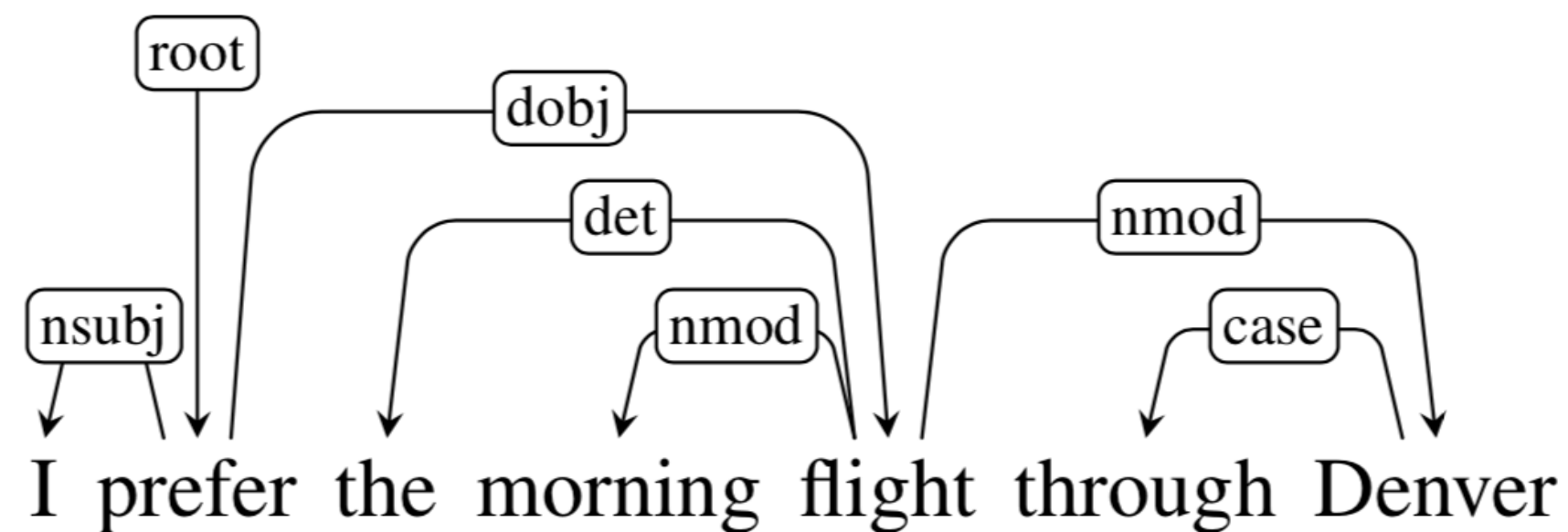
- Unlabeled attachment score (UAS)
= percentage of words that have been assigned the correct head
- Labeled attachment score (LAS)
= percentage of words that have been assigned the correct head & label



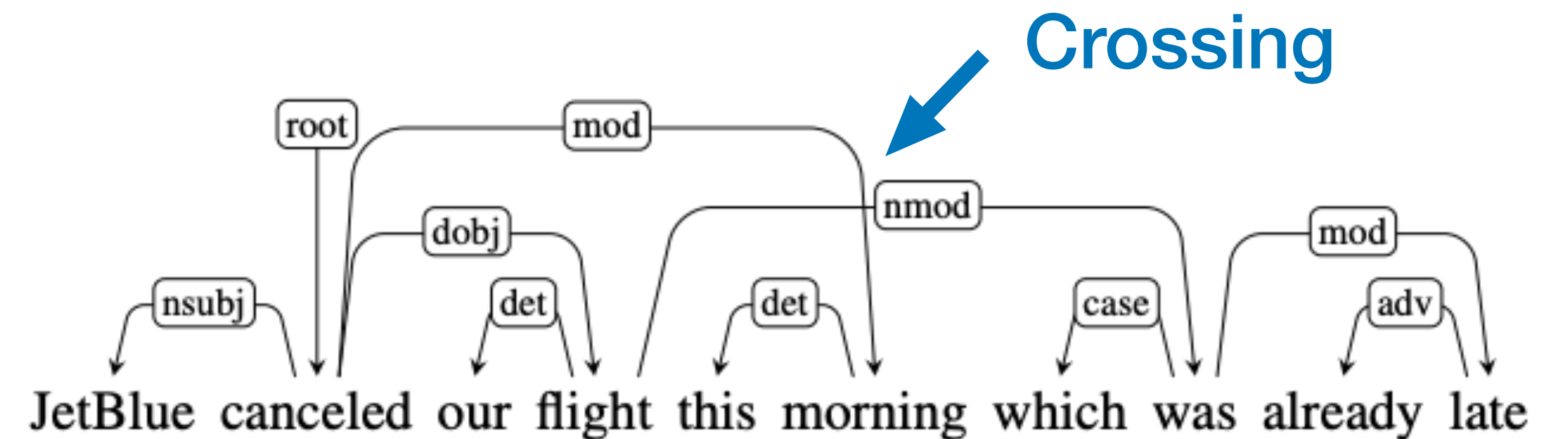
UAS = ? LAS = ?

Projectivity

- **Definition:** there are **no crossing dependency arcs** when the words are laid out in their linear order, with all arcs above the words



projective



non-projective

Non-projectivity arises due to long distance dependencies or in languages with flexible word order.

This class: focuses on projective parsing

Dataset	# Sentences	(%) Projective
English	39,832	99.9
Chinese	16,091	100.0
Czech	72,319	76.9
German	38,845	72.2

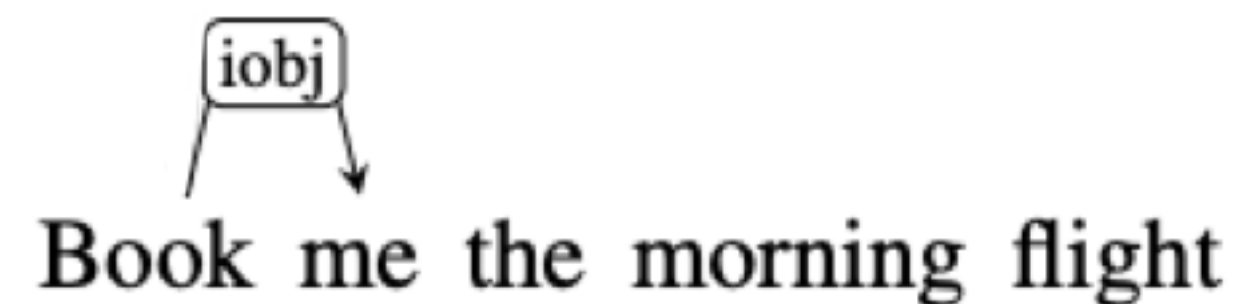
Transition-based dependency parsing

- The parsing process is modeled as a **sequence of transitions**
- A configuration consists of a **stack** s , a **buffer** b and a set of **dependency arcs** A : $c = (s, b, A)$

Stack: Can add arcs to 1st two words on stack

Buffer: Unprocessed words

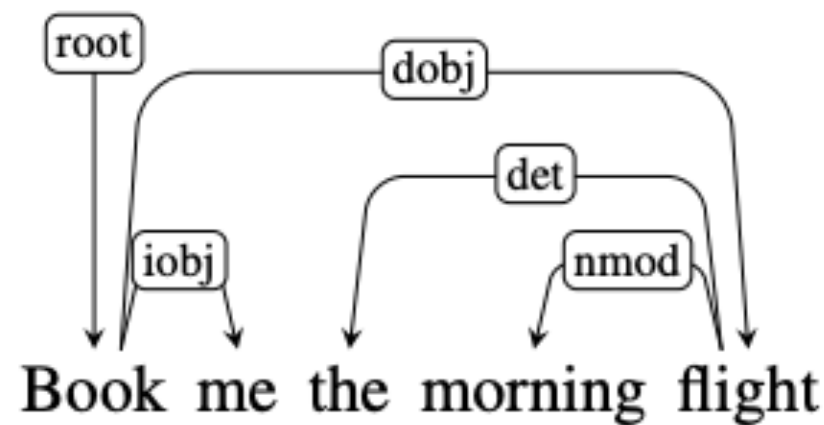
Current graph:



Transition-based dependency parsing

- The parsing process is modeled as a **sequence of transitions**
- A configuration consists of a **stack** s , a **buffer** b and a set of **dependency arcs** A : $c = (s, b, A)$
- Initially, $s = [\text{ROOT}]$, $b = [w_1, w_2, \dots, w_n]$, $A = \emptyset$
- Three types of transitions (s_1, s_2 : the top 2 words on the stack; b_1 : the first word in the buffer)
 - LEFT-ARC (r): add an arc $(s_1 \xrightarrow{r} s_2)$ to A , remove s_2 from the stack
 - RIGHT-ARC (r): add an arc $(s_2 \xrightarrow{r} s_1)$ to A , remove s_1 from the stack
 - SHIFT: move b_1 from the buffer to the stack
- A configuration is terminal if $s = [\text{ROOT}]$ and $b = \emptyset$

This is called “Arc-standard”; There are other transition schemes...

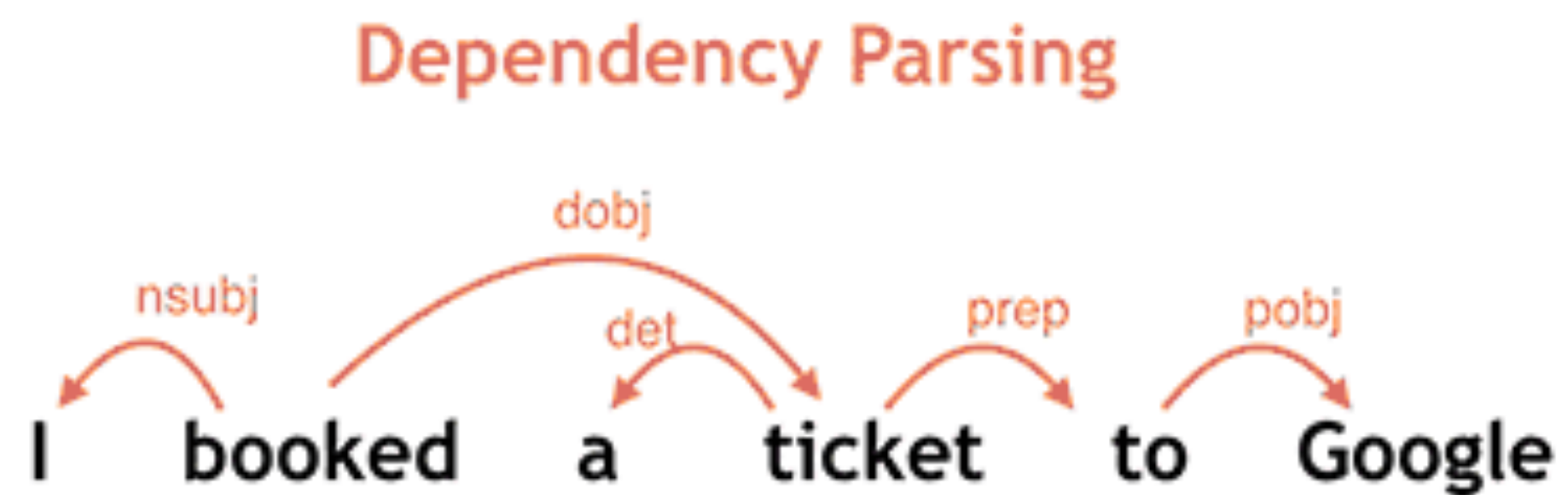


“Book me the morning flight”

A running example

	stack	buffer	action	added arc
0	[ROOT]	[Book, me, the, morning, flight]	SHIFT	
1	[ROOT, Book]	[me, the, morning, flight]	SHIFT	
2	[ROOT, Book, me]	[the, morning, flight]	RIGHT-ARC(iobj)	(Book, iobj, me)
3	[ROOT, Book]	[the, morning, flight]	SHIFT	
4	[ROOT, Book, the]	[morning, flight]	SHIFT	
5	[ROOT, Book, the, morning]	[flight]	SHIFT	
6	[ROOT, Book, the, morning, flight]	[]	LEFT-ARC(nmod)	(flight, nmod, morning)
7	[ROOT, Book, the, flight]	[]	LEFT-ARC(det)	(flight, det, the)
8	[ROOT, Book, flight]	[]	RIGHT-ARC(dobj)	(Book, dobj, flight)
9	[ROOT, Book]	[]	RIGHT-ARC(root)	(ROOT, root, Book)
10	[ROOT]	[]		

Transition-based dependency parsing

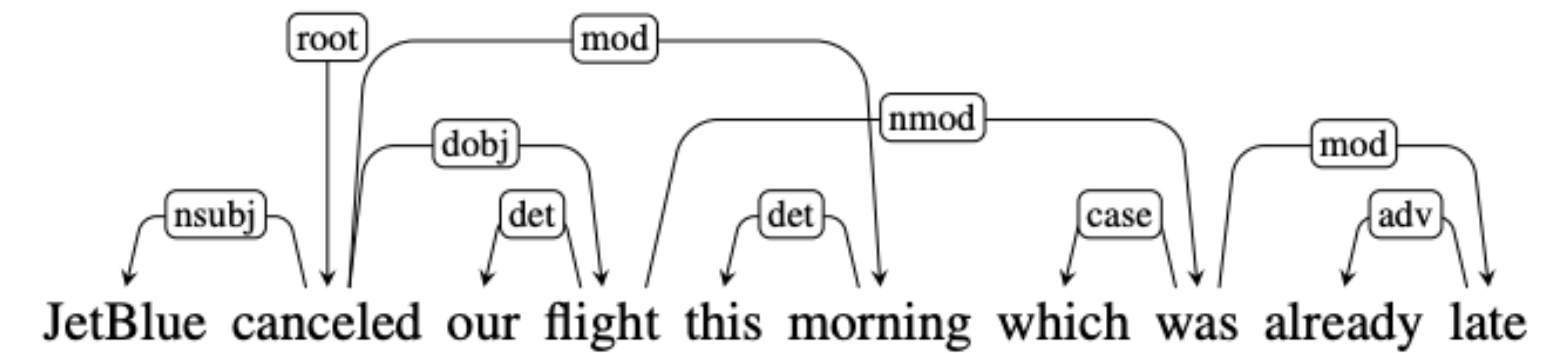


Transition-based dependency parsing

How many transitions are needed? How many times of SHIFT?

Correctness:

- For every complete transition sequence, the resulting graph is a projective dependency forest (**soundness**)
- For every projective dependency forest G, there is a transition sequence that generates G (**completeness**)
- However, one parse tree can have multiple valid transition sequences. **Why?**
 - “He likes dogs”
 - Stack = [ROOT He likes]
 - Buffer = [dogs]
 - Action = ??



Train a classifier to predict actions!

- Given $\{x_i, y_i\}$ where x_i is a sentence and y_i is a dependency parse
- For each x_i with n words, we can construct a transition sequence of length $2n$ which generates y_i , so we can generate $2n$ training examples: $\{(c_k, a_k)\}$ c_k : configuration, a_k : action
 - “shortest stack” strategy: prefer LEFT-ARC over SHIFT.

Given this information, the oracle chooses transitions as follows:

LEFTARC(r): **if** $(S_1 \ r \ S_2) \in R_p$

RIGHTARC(r): **if** $(S_2 \ r \ S_1) \in R_p$ **and** $\forall r', w \text{ s.t. } (S_1 \ r' \ w) \in R_p$ **then** $(S_1 \ r' \ w) \in R_c$

SHIFT: **otherwise**

- The goal becomes **how to learn a classifier from c_i to a_i**

How many training examples? How many classes?

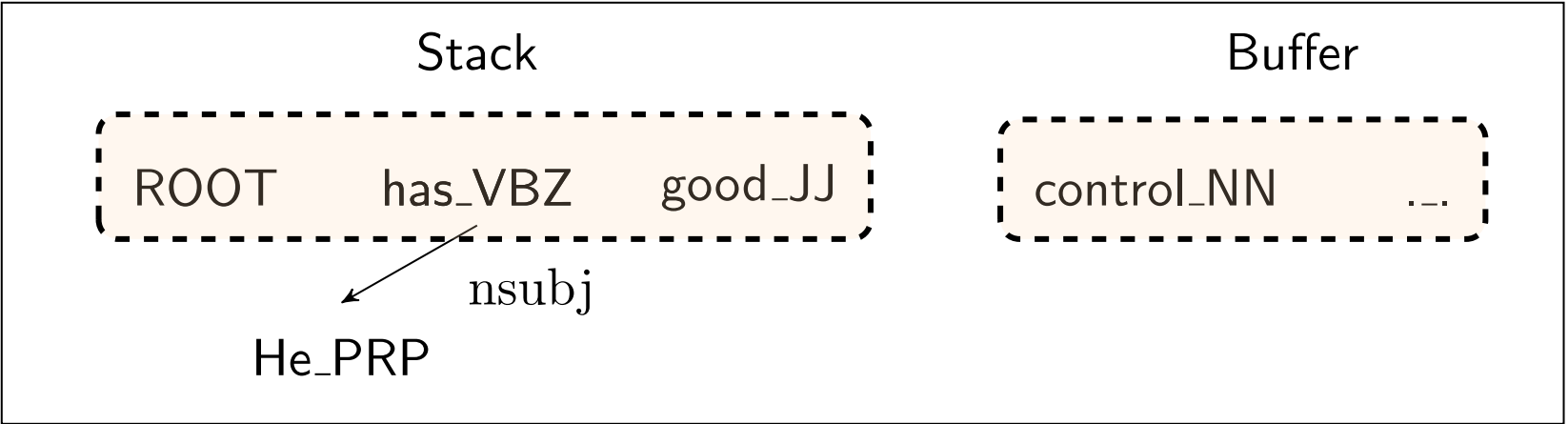
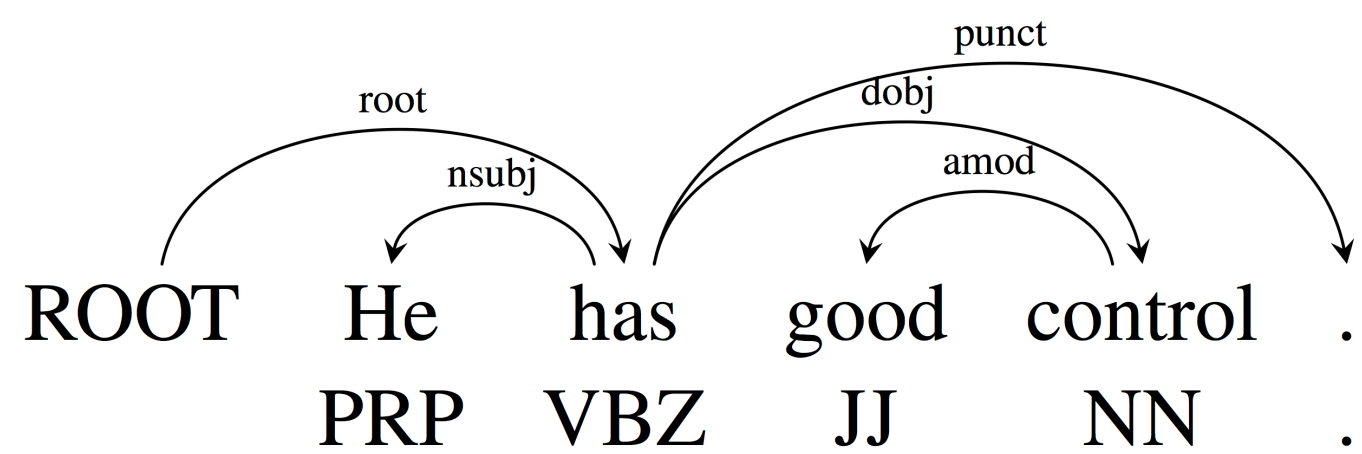
Train a classifier to predict actions!

- During testing, we use the classifier to repeat predicting the action, until we reach a terminal configuration

```
function DEPENDENCYPARSE(words) returns dependency tree  
  
state  $\leftarrow$  { [root], [words], [] } ; initial configuration  
while state not final  
    t  $\leftarrow$  Classifier (state) ; choose a transition operator to apply  
    state  $\leftarrow$  APPLY(t, state) ; apply it, creating a new state  
return state
```

- This is also called “greedy transition-based parsing” because we always make a local decision at each step
 - It is very fast (linear time!) but less accurate
 - Can easily do beam search

MaltParser

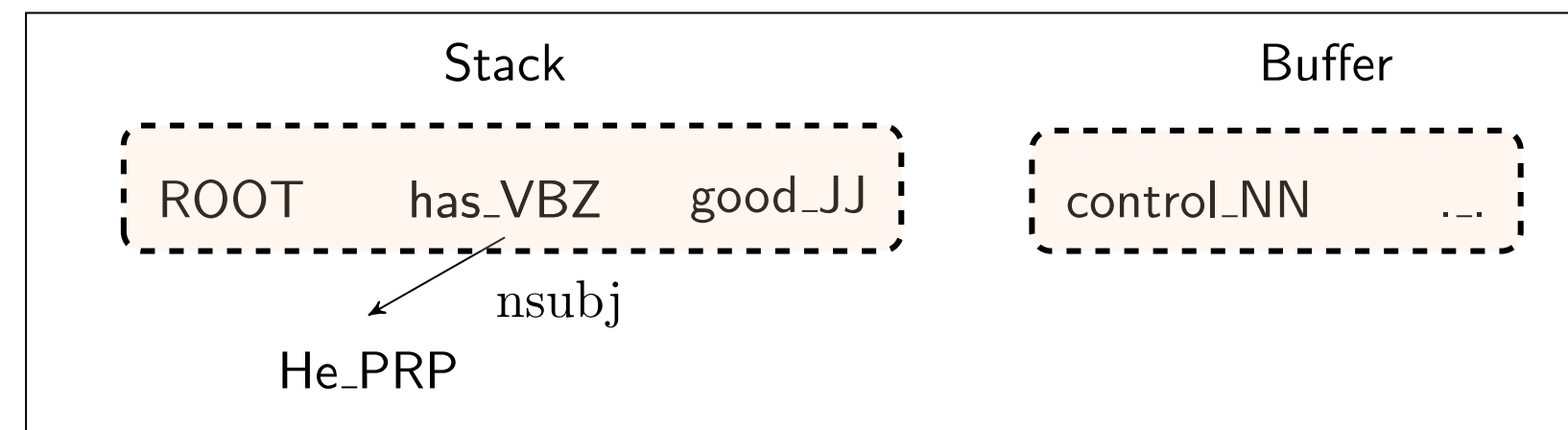


- Extract features from the configuration
- Use your favorite classifier: logistic regression, SVM...

Source	Feature templates		
One word	$s_1.w$	$s_1.t$	$s_1.wt$
	$s_2.w$	$s_2.t$	$s_2.wt$
	$b_1.w$	$b_1.w$	$b_0.wt$
Two word	$s_1.w \circ s_2.w$	$s_1.t \circ s_2.t$	$s_1.t \circ b_1.w$
	$s_1.t \circ s_2.wt$	$s_1.w \circ s_2.w \circ s_2.t$	$s_1.w \circ s_1.t \circ s_2.t$
	$s_1.w \circ s_1.t \circ s_2.t$	$s_1.w \circ s_1.t$	

w: word, t: part-of-speech tag

MaltParser



Feature templates

$$s_2 . w \circ s_2 . t$$

$$s_1 . w \circ s_1 . t \circ b_1 . w$$

$$lc(s_2) . t \circ s_2 . t \circ s_1 . t$$

$$lc(s_2) . w \circ lc(s_2) . l \circ s_2 . w$$

Features

$$s_2 . w = \text{has} \circ s_2 . t = \text{VBZ}$$

$$s_1 . w = \text{good} \circ s_1 . t = \text{JJ} \circ b_1 . w = \text{control}$$

$$lc(s_2) . t = \text{PRP} \circ s_2 . t = \text{VBZ} \circ s_1 . t = \text{JJ}$$

$$lc(s_2) . w = \text{He} \circ lc(s_2) . l = \text{nsubj} \circ s_2 . w = \text{has}$$

Usually a combination of 1-3 elements from the configuration

Binary, sparse, millions of features

More feature templates

```
# From Single Words
pair { stack.tag stack.word }
stack { word tag }
pair { input.tag input.word }
input { word tag }
pair { input(1).tag input(1).word }
input(1) { word tag }
pair { input(2).tag input(2).word }
input(2) { word tag }

# From word pairs
quad { stack.tag stack.word input.tag input.word }
triple { stack.tag stack.word input.word }
triple { stack.word input.tag input.word }
triple { stack.tag stack.word input.tag }
triple { stack.tag input.tag input.word }
pair { stack.word input.word }
pair { stack.tag input.tag }
pair { input.tag input(1).tag }

# From word triples
triple { input.tag input(1).tag input(2).tag }
triple { stack.tag input.tag input(1).tag }
triple { stack.head(1).tag stack.tag input.tag }
triple { stack.tag stack.child(-1).tag input.tag }
triple { stack.tag stack.child(1).tag input.tag }
triple { stack.tag input.tag input.child(-1).tag }

# Distance
pair { stack.distance stack.word }
pair { stack.distance stack.tag }
pair { stack.distance input.word }
pair { stack.distance input.tag }
triple { stack.distance stack.word input.word }
triple { stack.distance stack.tag input.tag }
```

```
# valency
pair { stack.word stack.valence(-1) }
pair { stack.word stack.valence(1) }
pair { stack.tag stack.valence(-1) }
pair { stack.tag stack.valence(1) }
pair { input.word input.valence(-1) }
pair { input.tag input.valence(-1) }

# unigrams
stack.head(1) {word tag}
stack.label
stack.child(-1) {word tag label}
stack.child(1) {word tag label}
input.child(-1) {word tag label}

# third order
stack.head(1).head(1) {word tag}
stack.head(1).label
stack.child(-1).sibling(1) {word tag label}
stack.child(1).sibling(-1) {word tag label}
input.child(-1).sibling(1) {word tag label}
triple { stack.tag stack.child(-1).tag stack.child(-1).sibling(1) }
triple { stack.tag stack.child(1).tag stack.child(1).sibling(-1) }
triple { stack.tag stack.head(1).tag stack.head(1).head(1).tag }
triple { input.tag input.child(-1).tag input.child(-1).sibling(1) }

# label set
pair { stack.tag stack.child(-1).label }
triple { stack.tag stack.child(-1).label stack.child(-1).sibling(1) }
quad { stack.tag stack.child(-1).label stack.child(-1).sibling(1) }
pair { stack.tag stack.child(1).label }
triple { stack.tag stack.child(1).label stack.child(1).sibling(-1) }
quad { stack.tag stack.child(1).label stack.child(1).sibling(-1) }
pair { input.tag input.child(-1).label }
triple { input.tag input.child(-1).label input.child(-1).sibling(1) }
quad { input.tag input.child(-1).label input.child(-1).sibling(1) }
```

Parsing with neural networks

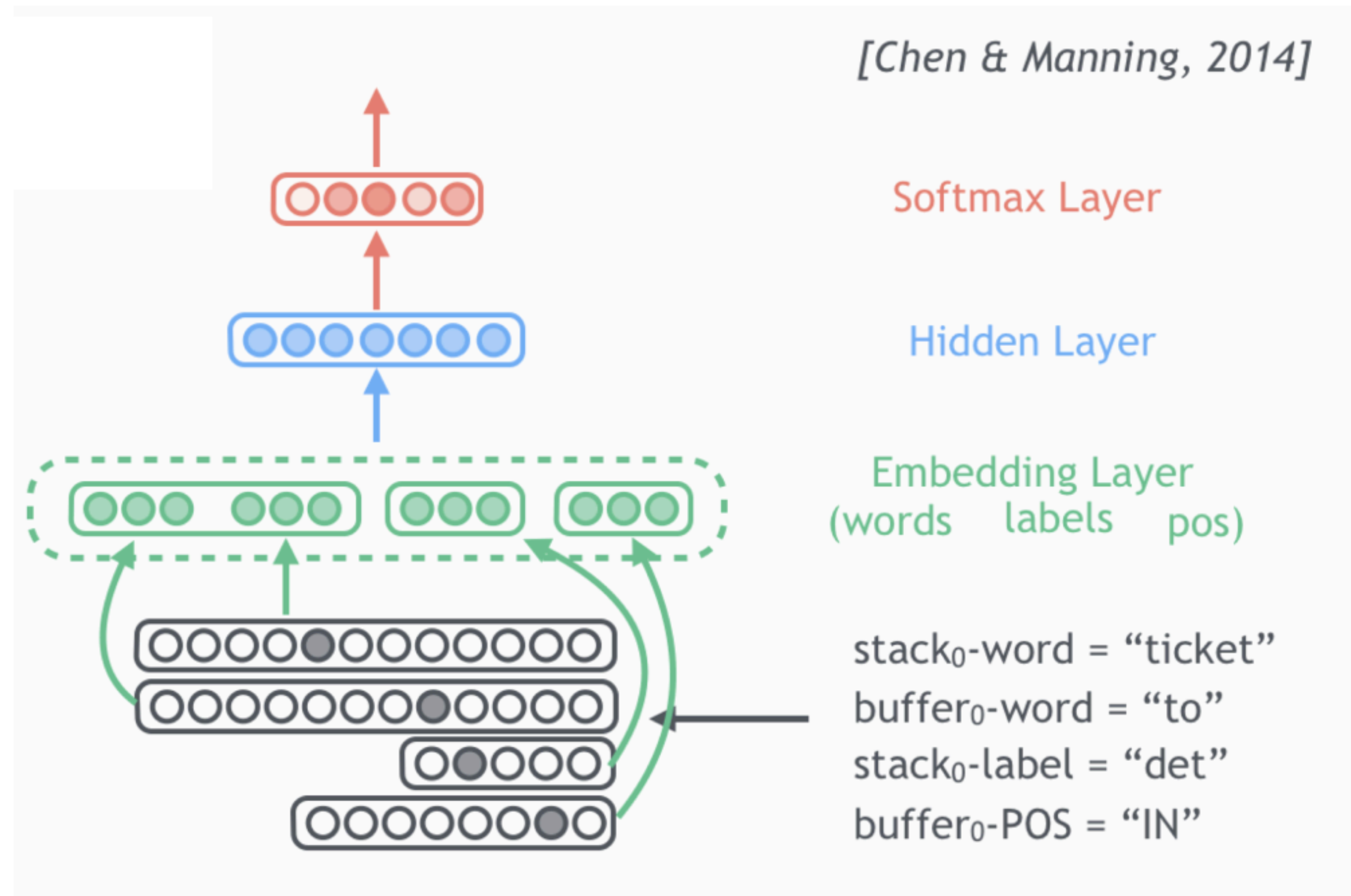


Representation for configuration:

- Embeddings for words/POS tags on top of stack
- Embeddings for words/POS tags at front of buffer
- Embeddings for existing arc labels at specific positions

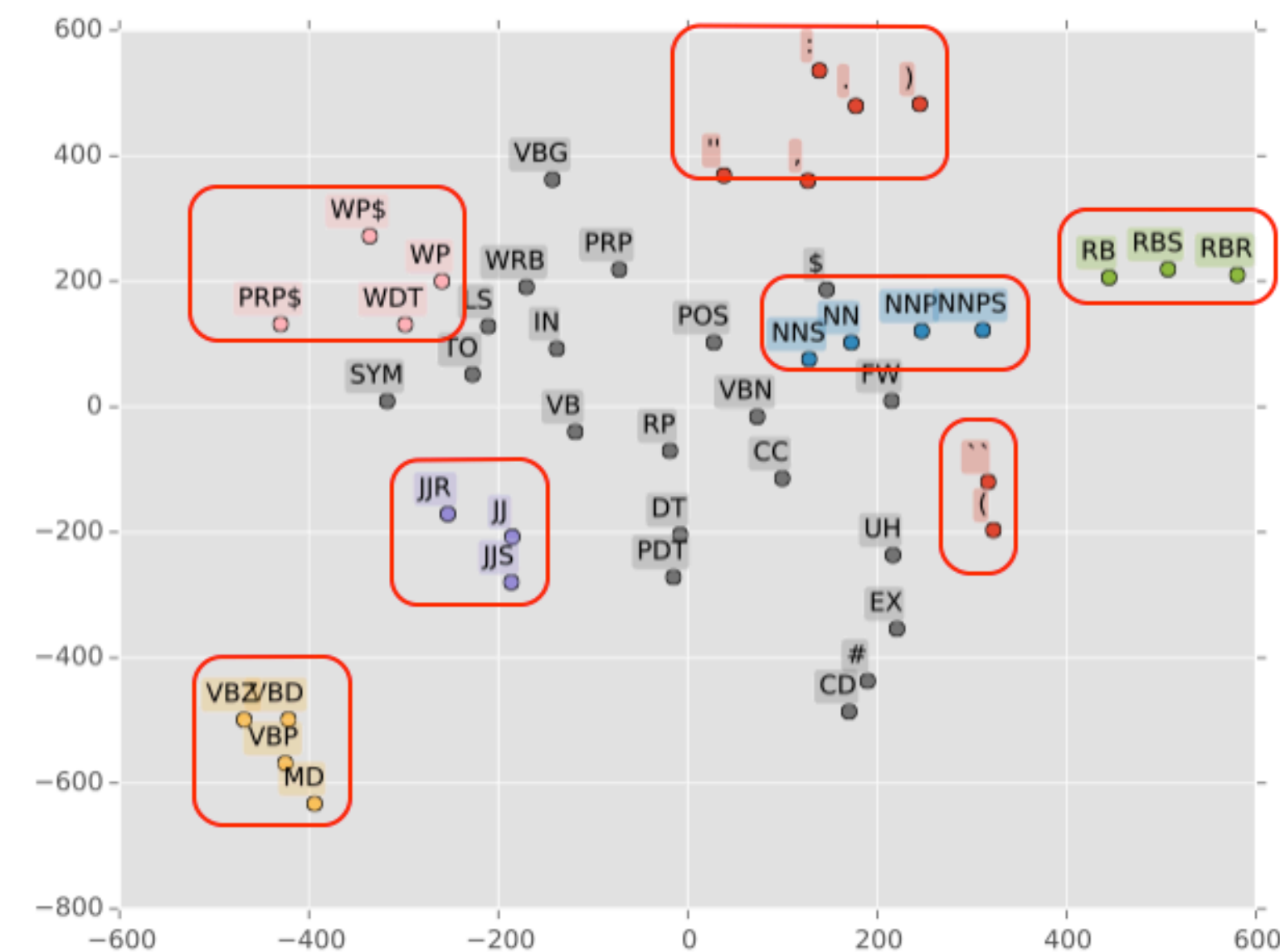
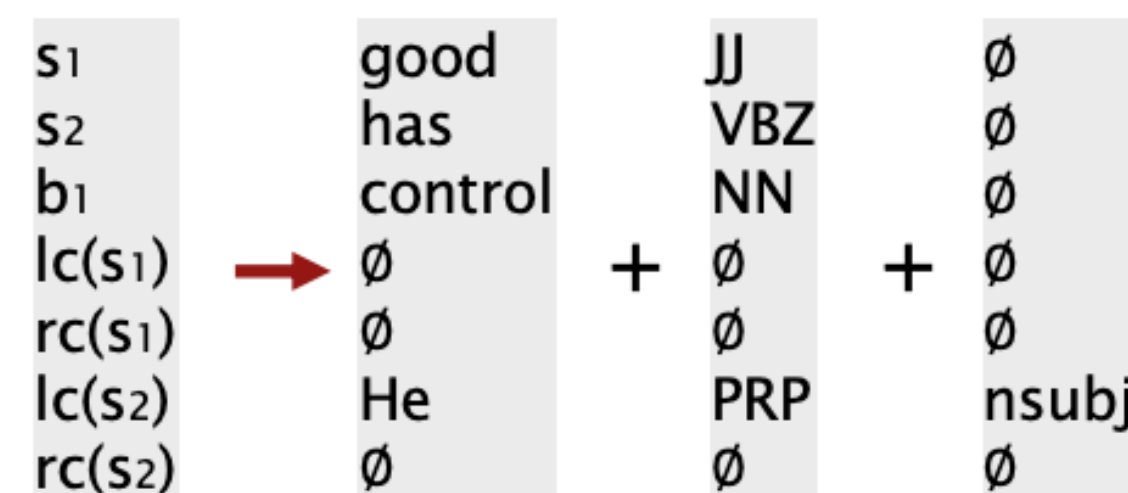
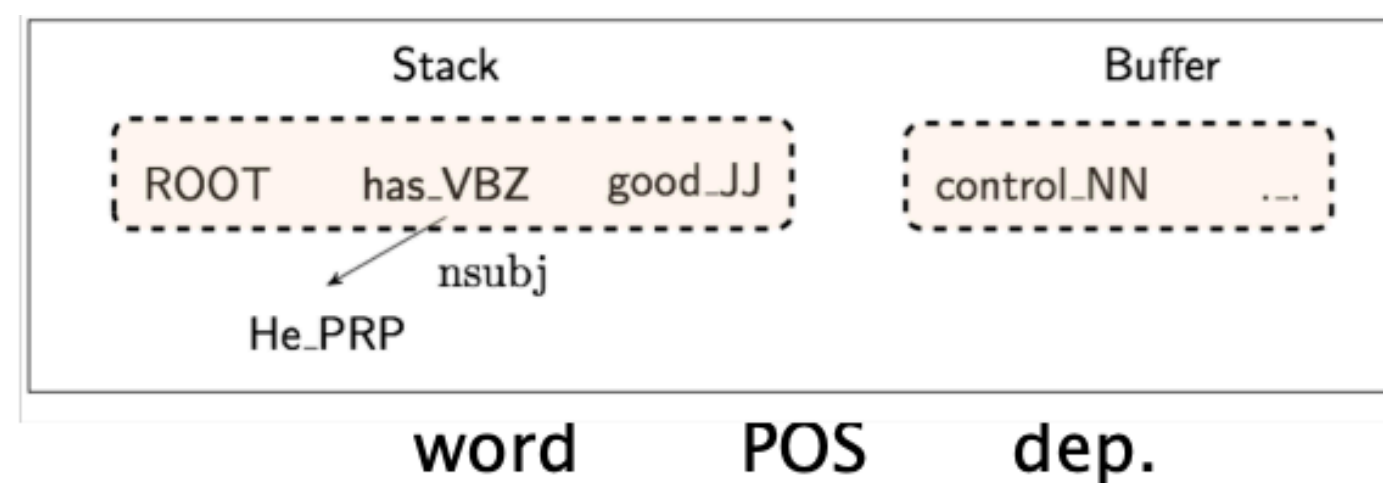
Classifier:

- Feed-forward neural network (input representation has a fixed dimensionality)



Parsing with neural networks

- Used pre-trained word embeddings
- Part-of-speech tags and dependency labels are also represented as vectors
- No feature template any more!



Parser	UAS	LAS	sent. / s
MaltParser	89.8	87.2	469
MSTParser	91.4	88.1	10
TurboParser	92.3	89.6	8
C & M 2014	92.0	89.7	654

- A simple feedforward NN: what is left is backpropagation!

(Chen and Manning, 2014): A Fast and Accurate Dependency Parser using Neural Networks

Further improvements

- Bigger, deeper networks with better tuned hyperparameters
- Beam search
- Global normalization

Method	UAS	LAS (PTB WSJ SD 3.3)
Chen & Manning 2014	92.0	89.7
Weiss et al. 2015	93.99	92.05
Andor et al. 2016	94.61	92.79

Google's SyntaxNet and the Parsey McParseFace (English) model

Announcing SyntaxNet: The World's Most Accurate Parser
Goes Open Source

Thursday, May 12, 2016

Handling non-projectivity

- The arc-standard algorithm we presented only builds projective dependency trees
- Possible directions:
 - Give up!
 - Post-processing
 - Add new transition types (e.g., SWAP)
 - Switch to a different algorithm (e.g., graph-based parsers such as MSTParser)

Graph-based dependency parsing

- **Basic idea:** let's predict the dependency tree directly

$$Y^* = \arg \max_{Y \in \Phi(X)} \text{score}(X, Y)$$

X: sentence, Y: any possible dependency tree

- **Factorization:**

edges

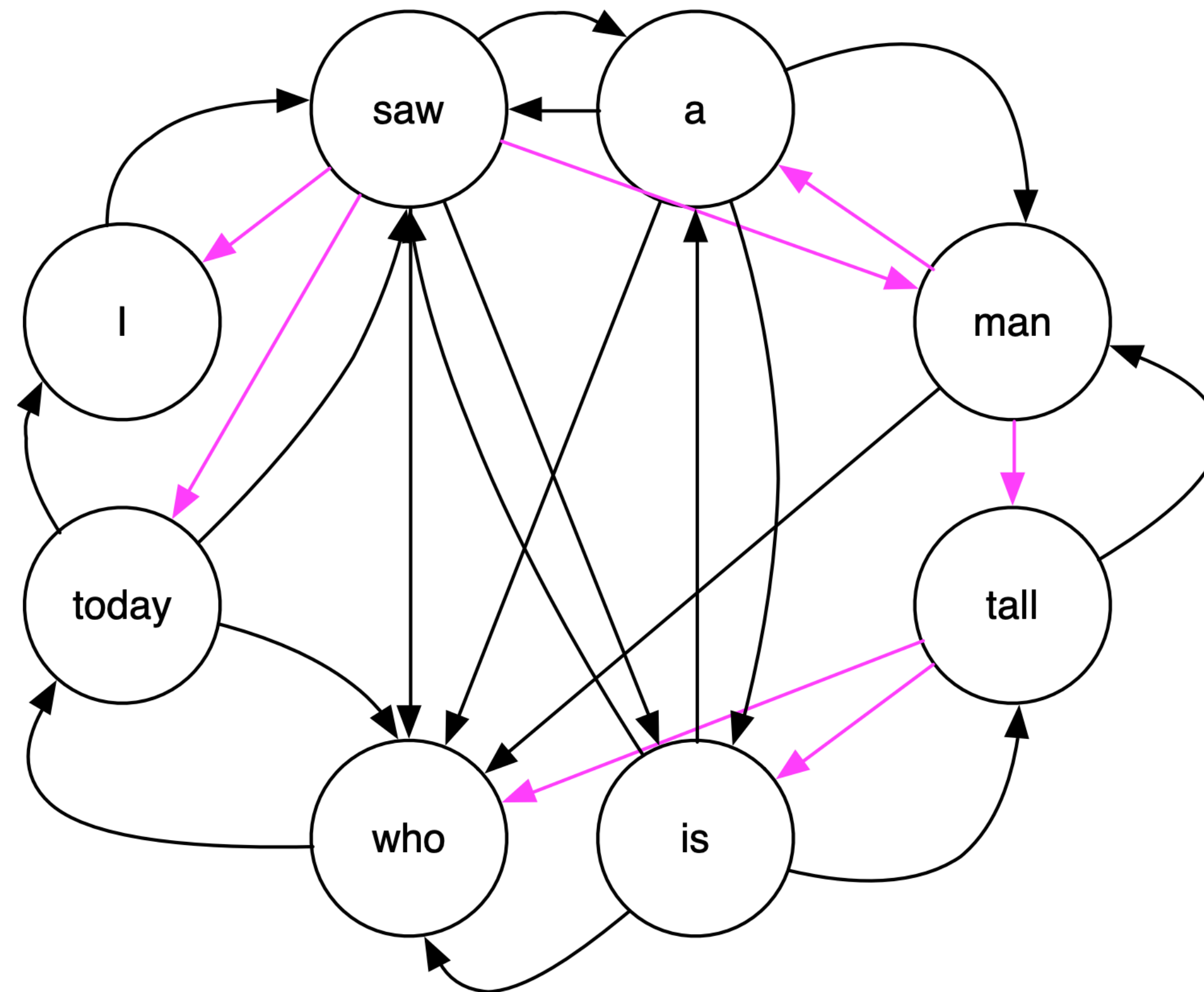
$$\text{score}(X, Y) = \sum_{e \in Y} \text{score}(e) = \sum_{e \in Y} w^T f(e)$$

Use a neural network to compute these scores

- **Inference:** finding maximum spanning tree (MST) for weighted, directed graph

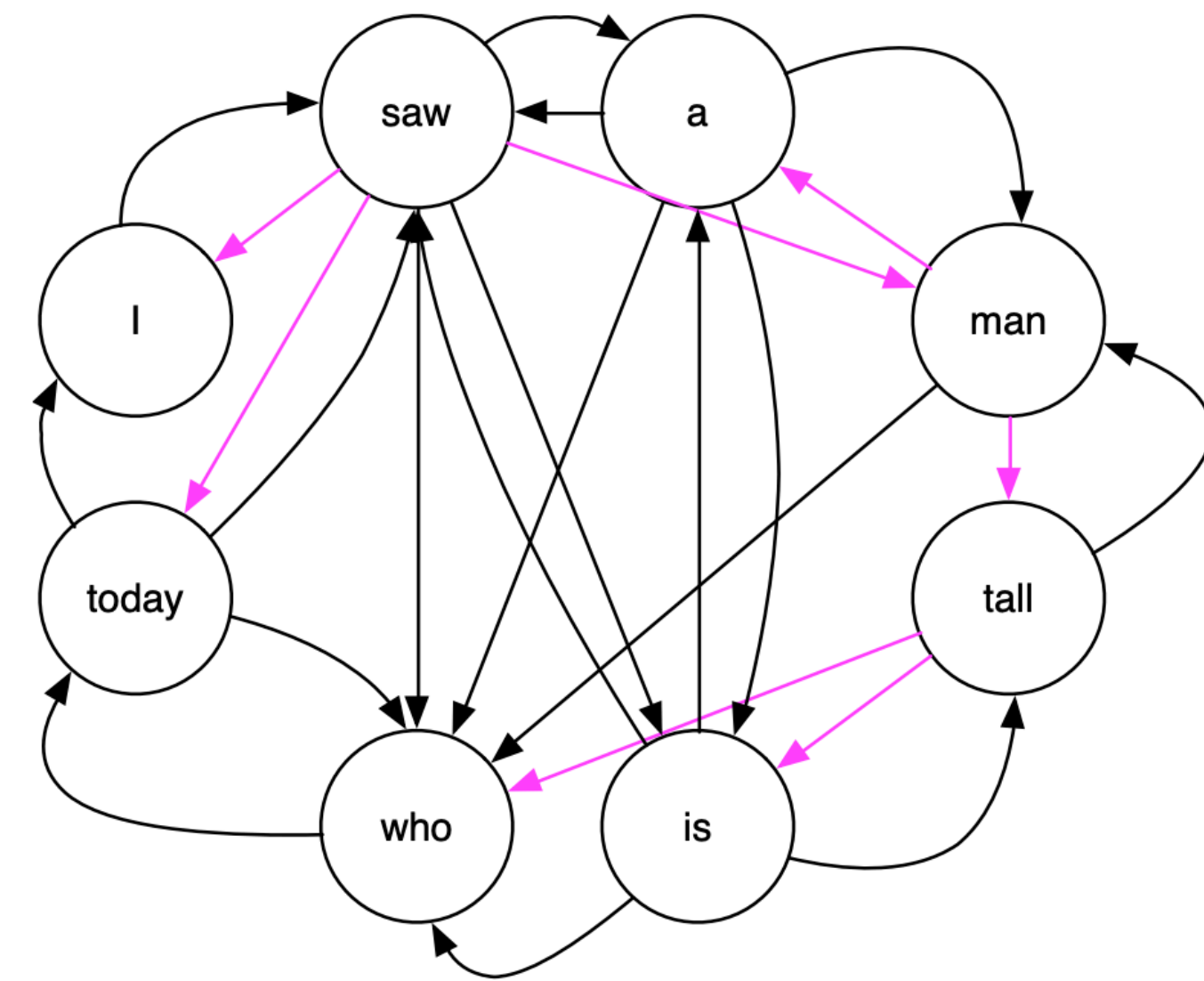
MST Parsing Inference

- We start out with a fully connected graph with a score for each edge
- N^2 edges total



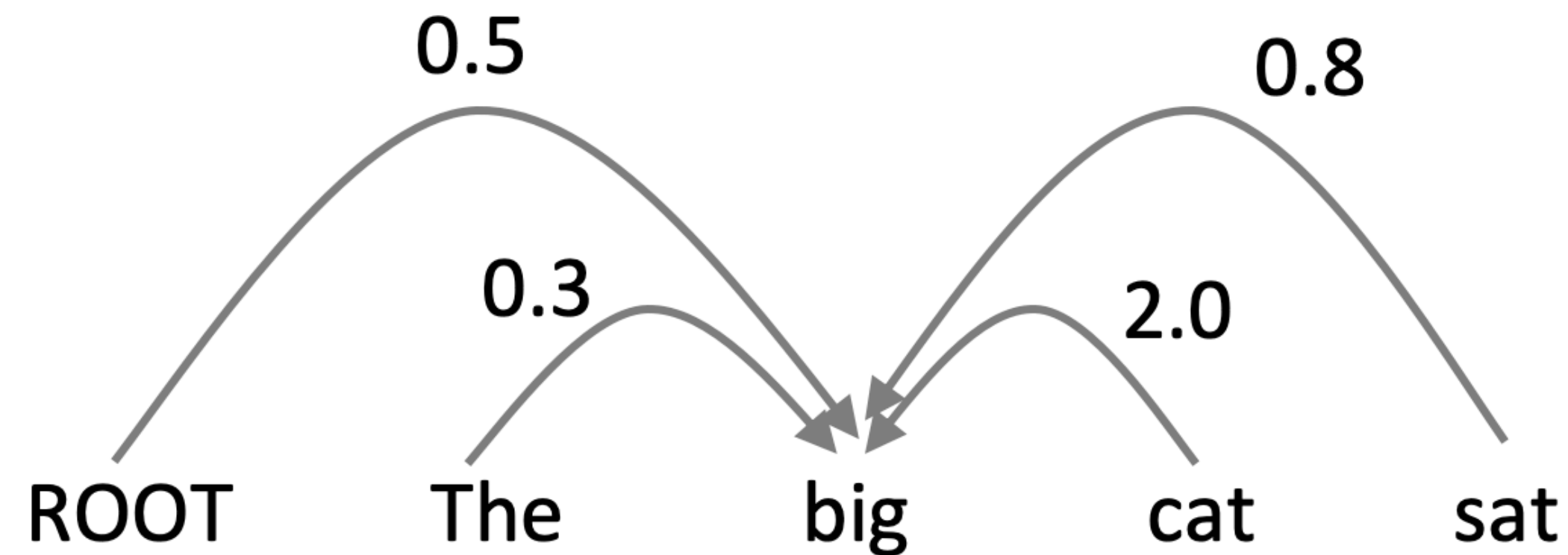
MST Parsing Inference

- From this graph G , we want to find a **spanning tree** (tree that spans G [includes all the vertices in G])
- If the edges have weights, the best parse is the **maximal spanning tree** (the spanning tree with the highest total weight).



Graph-based dependency parsing

- **Training** learn parameters so the score for the gold tree is higher than for all other trees
- **Compute** a score for every possible dependency for each word using good “contextual” representations of each word token

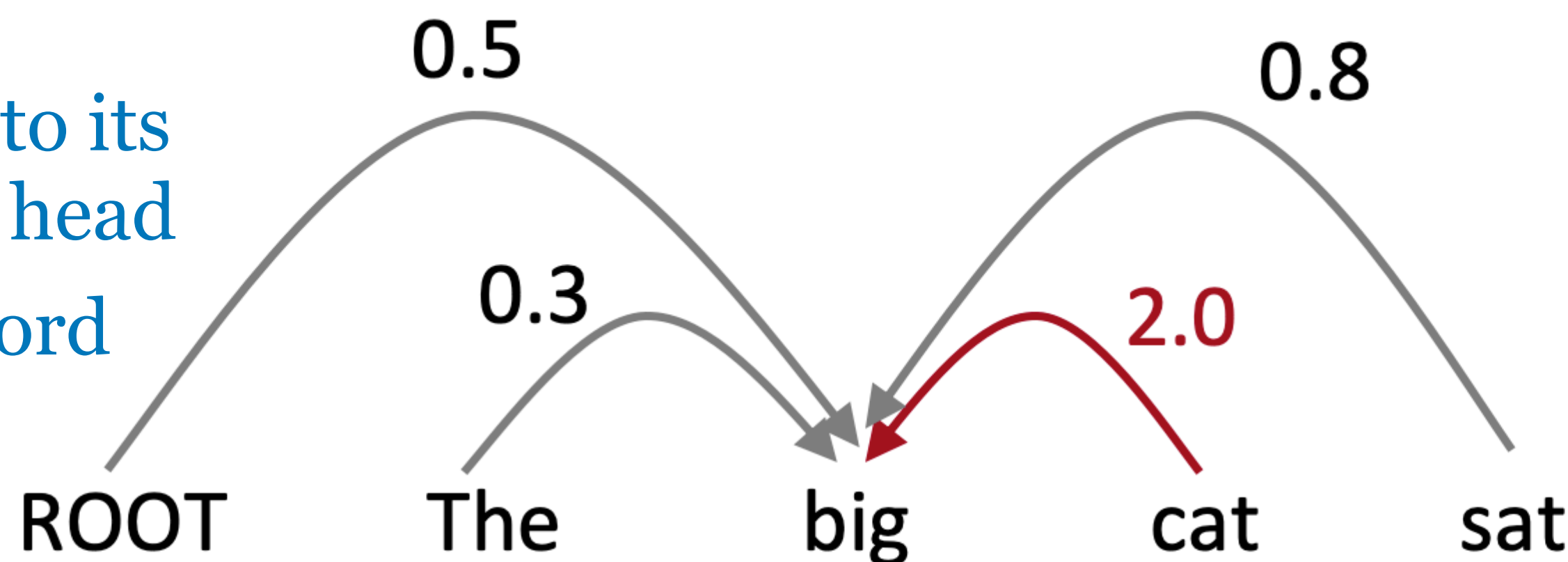


e.g., picking the head for “big”

Graph-based dependency parsing

- **Training** learn parameters so the score for the gold tree is higher than ~~for all other trees~~ **a single best tree**
- **Compute** a score for every possible dependency for each word using good “contextual” representations of each word token

- Add edge from each word to its highest-scoring candidate head
- Repeat process for each word



e.g., picking the head for “big”

Neural graph-based dependency parser (Dozat and Manning 2017)

- Great result!
- But slower than simple neural transition-based parsers
 - There are n^2 possible dependencies in a sentence of length n

Method	UAS	LAS (PTB WSJ SD 3.3)
Chen & Manning 2014	92.0	89.7
Weiss et al. 2015	93.99	92.05
Andor et al. 2016	94.61	92.79
Dozat & Manning 2017	95.74	94.08